

Lyrus Life Sciences: Comprehensive Business Intelligence and Technical Research Report

1. Corporate History and Financial Architecture

1.1 Genesis, Corporate Evolution, and the Medreich Legacy

The contemporary landscape of the global pharmaceutical contract services industry is characterized by the rapid evolution of specialized Contract Research and Development Organizations (CRDOs). Within this highly technical domain, Lyrus Life Sciences Private Limited has established itself as a formidable entity. Headquartered in Bengaluru, Karnataka, India, the enterprise operates at the critical intersection of proprietary complex generic formulation, contract research, and specialized global pharmaceutical trading. To fully comprehend the operational posture and strategic mandate of Lyrus Life Sciences, it is imperative to analyze its corporate genesis, which is inextricably linked to the legacy of its primary promoter, Mr. C.P. Bothra.

Mr. Bothra is a highly distinguished veteran of the Indian pharmaceutical sector, commanding over four decades of intensive industry experience.¹ His foundational success was cemented as the Founder and Managing Director of Medreich Limited, a pioneering Contract Research and Manufacturing Services (CRAMS) organization established in 1976.² Under his stewardship, Medreich evolved from a nascent outsourced manufacturing partner for global giants like GlaxoSmithKline into a fully integrated global pharmaceutical powerhouse.³

The defining moment of this legacy occurred in 2015 when Medreich was successfully acquired by the Japanese pharmaceutical conglomerate Meiji Seika Pharma Co., Ltd..² This acquisition represented a massive liquidity event; industry financial analyses indicate that Mr. Bothra received an estimated sum exceeding ₹250 Crore from the sale of his equity stake in the Medreich venture.⁵

Leveraging this substantial infusion of liquid capital, Lyrus Life Sciences Private Limited was formally incorporated on June 16, 2015, with active, scaled operations commencing in 2016.⁶ Rather than relying on external venture capital or private equity syndication, the organization operates as a promoter-funded entity. The capital from the Medreich exit was strategically channeled into a holding architecture, specifically a partnership firm registered as Nokha Trading LLP, which serves as the primary funding conduit and holding company for Lyrus.⁵ From 2016 to the present day, Lyrus has executed a highly ambitious, dual-pronged business model. The company simultaneously manages a capital-intensive, high-risk contract research and development division alongside a profitable, high-volume pharmaceutical trading operation focused entirely on highly regulated export markets.⁵

Over the past decade, the organization has systematically expanded its global footprint, eschewing the fragmented domestic Indian retail market in favor of stringent regulatory environments. This strategy has resulted in the successful registration of over 60 commercial products and 150 Stock Keeping Units (SKUs) across more than twenty international jurisdictions, including the United Kingdom, Australia, the European Union, and South Africa.⁴

A landmark achievement validating this regulatory strategy occurred in early 2023 when Lyrus filed its inaugural Abbreviated New Drug Application (ANDA) for the highly competitive United States market, culminating in a rare, first-cycle approval by the US FDA in December 2023 for its generic urological intervention, Methenamine Hippurate.⁹

1.2 Executive Leadership and Human Capital

The strategic direction and operational execution at Lyrus Life Sciences are governed by a specialized leadership consortium that brings together over 300 years of cumulative experience in pharmaceutical formulation, cGMP manufacturing compliance, and global commercialization.¹

As of mid-2024, the enterprise maintains an estimated overall headcount of 107 full-time employees, reflecting a measured 9% year-over-year growth trajectory.⁶ This headcount includes an elite technical workforce of over 65 specialized research scientists dedicated exclusively to the R&D division.⁵

The organizational hierarchy is designed to blend deep pharmaceutical science with rigorous financial governance. Table 1 provides an exhaustive profile of the key executives steering the enterprise.

Executive Name	Current Designation	Background, Industry Affiliation, and Strategic Role
C.P. Bothra	Chairman	Over 47 years in the pharmaceutical business; former Managing Director and co-owner of Medreich Ltd. He also serves as the Chairman of Stabicon Life Sciences and Trust Chemists and Druggists. Furthermore, he is the Managing Trustee of the Bothra Foundation, managing philanthropic healthcare initiatives for over 20,000 patients. ²
Hemanth Kumar Bothra	Managing Director & Co-Founder	A core promoter who has managed operations since the company's inception. He concurrently leads the operations of the South Indian pharmaceutical retail chain, Trust Chemists and Druggists. ⁶
Leela Kanwar	Co-Founder & Director	A central promoter and strategic director associated with the overarching corporate architecture. She holds directorships in Nokha Trading LLP, QP Health Services Limited, and Equidor Medtech LLP. ¹⁶
Rajesh Kumar Goel	Chief Executive Officer (CEO)	A highly seasoned corporate governance and finance professional holding triple professional degrees as a Chartered Accountant (CA), Company Secretary (CS), and Cost and Works Accountant (ICWA). Interestingly, his background includes entrepreneurial diversification as the co-owner of the Spaces Spa chain in Karnataka, demonstrating versatile operational acumen. ⁶
Uma Maheshwari	VP - Global Compliance	Brings 24 years of rigorous pharmaceutical industry experience spanning Quality Assurance, Regulatory Affairs, pharmacovigilance, and complex technology transfers. She holds a Bachelor's degree in Pharmacy alongside a Master's degree in Pharmaceutical Technology, ensuring strict compliance with FDA and EMA mandates. ⁹
Elayaraja Natarajan	Head of R&D	A leading formulation scientist with over 15 years of deep product development expertise. His previous industry affiliations include top-tier CDMOs such as Strides Shasun, Rubicon Research, Genovo, and Medreich. He has been instrumental in the architectural design of hundreds of ANDAs and Marketing Authorizations for the US and European markets. ⁹

1.3 Capital Structure, Financial Performance, and Macroeconomic Metrics

The financial architecture of Lyrus Life Sciences perfectly encapsulates the fundamental economic tension inherent in the CRDO business model. The enterprise relies on its high-volume pharmaceutical trading operations to generate baseline revenue and operational profitability; however, the capital-intensive R&D division consistently operates at a profound net loss due to the massive, continuous expenditure required for proprietary formulation development, clinical bioequivalence trials, and multi-jurisdictional regulatory filings.⁵

Lyrus maintains an authorized share capital of ₹96.25 Crore, alongside a nearly fully utilized paid-up capital base of ₹96.23 Crore.⁸ Because the firm's strategic mandate prioritizes the long-term accumulation of intellectual property over short-term net profit, the persistent operational deficits generated by the R&D burn rate are consistently bridged by direct capital injections from the promoters via the Nokha Trading LLP

holding structure.⁵

Table 2 delineates the standalone financial performance metrics of Lyrus Life Sciences across recent fiscal cycles.

Financial Metric (₹ in Crores)	FY 2021 (Audited)	FY 2022 (Audited)	FY 2023 (Provisional)	FY 2025 (Estimated)
Total Operating Income	76.72	76.17	67.71	60.90
EBITDA	-1.05	3.02	2.24	N/A
EBITDA Margin (%)	-1.37%	3.96%	3.31%	N/A
Profit After Tax (PAT)	-6.32	-5.79	-4.49	N/A
Total Debt	75.53	91.20	104.52	N/A
Adjusted Tangible Net Worth	75.16	63.48	70.66	N/A
Overall Gearing Ratio (x)	0.88	1.25	1.16	N/A

Analytical Note: Data derived from verified Infomerics credit rating reports and corporate intelligence databases.⁵

An analysis of these financial metrics reveals a complex operational narrative. While the company achieved a total operating income of ₹76.72 Crore in FY21, this figure has demonstrated a steady contraction, dropping to ₹67.71 Crore in FY23 (Provisional) and further descending to an estimated ₹60.9 Crore by the conclusion of FY25, representing a negative Compound Annual Growth Rate (CAGR) of approximately 22%.⁵ This revenue contraction is likely attributable to the unpredictable timing of milestone-based R&D revenue recognition, alongside potential supply chain realignments in its core export markets.

Despite generating positive operational margins at the EBITDA level during FY22 and FY23, the net bottom line (Profit After Tax) remains firmly negative. This dynamic is a direct consequence of the aggressive financing required to build out the Hoskote R&D infrastructure and fund the formulation pipeline. To sustain this trajectory, Lyrus has heavily leveraged its balance sheet, with total debt escalating significantly from ₹75.53 Crore in FY21 to ₹104.52 Crore in FY23.²⁴ Consequently, the overall gearing ratio has surpassed 1.0x, indicating a capital structure that is increasingly reliant on debt financing alongside promoter equity to maintain its aggressive operational tempo.⁵

2. Specialized Capabilities, R&D Infrastructure, and Niche Technologies

2.1 Comprehensive CRDO Value Chain Integration

The operational core of Lyrus Life Sciences is centralized within a highly sophisticated, 75,000 square-foot research and development facility located in the Hoskote Industrial Area on the outskirts of Bengaluru, Karnataka.⁹ This facility serves as the nerve center for the firm's contract R&D division and houses a dedicated staff of over 90 personnel, with the infrastructure possessing the spatial capacity for a further 30,000 square-foot expansion to accommodate future technological integration.⁹

The scientific architecture of the Hoskote facility is meticulously segmented into highly specialized functional teams designed to shepherd a pharmaceutical molecule from nascent ideation through to final commercialization. The Ideation Team is tasked with the identification of prospective drug candidates, evaluating the patent landscape for non-infringing formulation pathways, and conducting first-level feasibility analyses on complex therapeutic targets.⁵

Once a molecule is selected, it transitions to the Pre-formulation and Formulation Development teams. These scientists conduct rigorous physicochemical profiling of active pharmaceutical ingredients (APIs), assess excipient compatibilities, and engineer complex dosage architectures. Lyrus possesses the technical

capability to develop a vast array of dosage forms, including conventional oral solid dosages (tablets and capsules), advanced soft gels, highly specialized extended-release suspensions, multi-unit pellet systems, topical ointments, emulgels, liquid orals, sterile injectables, and ophthalmics.⁵

Following physical formulation, the molecules are subjected to the Analytical Development division. Operating strictly under current Good Manufacturing Practices (cGMP) and International Council for Harmonisation (ICH) guidelines, this team provides comprehensive analytical support for all phases of drug development.²⁷ Their capabilities include the development and validation of stability-indicating assays, dissolution profiling, and generating the immense datasets required for regulatory submission.⁵ Finally, the Regulatory Affairs team synthesizes this laboratory data to compile comprehensive product dossiers, manage the complete lifecycle of the application, and draft highly technical responses to queries from stringent regulatory authorities such as the FDA and EMA.⁵

2.2 Global Manufacturing Alliances and cGMP Compliance

In contrast to traditional pharmaceutical manufacturers that sink massive amounts of capital into rigid, large-scale production facilities, Lyrus Life Sciences employs an agile, asset-light manufacturing strategy. The company maintains an in-house pilot plant specifically purposed for technical scale-up studies, the execution of critical validation batches, and the production of clinical exhibit materials necessary for bioequivalence testing.⁹

For commercial-scale manufacturing, Lyrus has strategically engaged in capacity-utilization alliances with eleven top-tier Contract Manufacturing Organizations (CMOs) globally.⁹ This decentralized manufacturing paradigm provides Lyrus with distinct competitive advantages. It guarantees an undivided corporate focus on the highly profitable core competency of formulation development while providing absolute portability.²⁸ Because Lyrus develops niche molecules—ranging from sterile injectables to complex emulgels—it frequently requires manufacturing capabilities that extend beyond the technological limitations or economic viability of domestic Indian facilities. By outsourcing to specialized global CMOs, Lyrus ensures that every product is manufactured in the optimal environment.²⁸

Crucially, every single contracted manufacturing site within the Lyrus network is rigorously vetted and holds authoritative approvals from the most stringent global regulatory bodies. These certifications include the US FDA, the United Kingdom's MHRA, the European Union's EU GMP framework, Australia's TGA, and the globally recognized PIC/S standard.⁴ This unbroken chain of cGMP compliance ensures that the dossiers developed by Lyrus face minimal friction during the commercial approval process in heavily regulated western markets.

2.3 Deep Dive: Novel Drug Delivery Systems (NDDS) and Nanotechnology

The true technical differentiator that elevates Lyrus Life Sciences above standard generic manufacturers is its profound expertise in Novel Drug Delivery Systems (NDDS) and bioavailability enhancement technologies. A massive proportion of modern pharmaceutical compounds, particularly those classified under the Biopharmaceutics Classification System (BCS) as Class II or Class IV, suffer from poor aqueous solubility, which severely limits their therapeutic efficacy. Lyrus's formulation scientists deploy advanced engineering methodologies to overcome these critical pharmacokinetic limitations.²⁶

A paramount example of this capability is the company's utilization of advanced nanotechnology to exponentially increase the oral bioavailability and transdermal penetration of highly hydrophobic compounds. This expertise is explicitly detailed in their work with complex nutraceuticals and standardized herbal extracts, such as Turmeric Oleoresin (TO) and highly purified curcumin powders. In their native state, these raw materials possess large, insoluble particle sizes averaging around 2 micrometers (μm), which the human gastrointestinal tract struggles to absorb effectively.³⁰

To solve this, Lyrus subjects these compounds to a proprietary nanonization process. By employing intense mechanical shear and advanced solvent-engineering techniques, the formulation teams successfully reduce the starting material down to an ultra-fine, highly stable nanoparticle range of 40 to 157 nanometers (nm).³⁰

At this nanoscale, the surface area to volume ratio of the compound increases dramatically, leading to a massive surge in dissolution rates and systemic absorption. To rigorously quantify, track, and validate this nanometric scaling, the Hoskote facility utilizes advanced backscattering analytical instrumentation, specifically relying on Brookhaven Nano analyzers. These highly sensitive instruments measure particle size and zeta potential by building complex mathematical correlation functions based on the dynamic light scattering of the nanoparticles suspended in solution.³⁰

Beyond physical nanonization, Lyrus excels in complex base modifications and solubility architecture. Traditional methods of forcing hydrophobic drugs into aqueous solution often require harsh pH buffering agents, which can cause severe gastrointestinal irritation or destabilize the active ingredient over its shelf life. Lyrus has successfully engineered water-soluble formulations of curcuminoids that remain entirely stable and therapeutically safe without the inclusion of any pH buffers, a technological leap that forms the basis of their granted US patent architecture.³² This ability to create non-infringing patent technologies around complex generic molecules is highly lucrative, allowing Lyrus to approach major pharmaceutical entities for joint development contracts and highly profitable sales and distribution arrangements.⁹

3. Product Portfolio, Therapeutic Segments, and Pipeline Commercialization

3.1 Commercialized Portfolio and Strategic Therapeutic Focus

Lyrus Life Sciences commands a highly diversified formulation portfolio that strategically straddles multiple distinct market segments, including ethical prescription therapeutics (both branded and generic), specialty nutraceuticals, and high-volume Over-the-Counter (OTC) products.¹⁰ The company has deliberately engineered its product mix to address the surging global demand for interventions in chronic and critical disease states.²⁷

The organization's therapeutic focus is compartmentalized into several key segments, each demanding specific formulation expertise. In the realm of the Central Nervous System (CNS) and Neuropsychiatry, Lyrus is actively developing molecules that require challenging delivery systems capable of achieving predictable blood-brain barrier penetration.²⁷ Their Cardiovascular System portfolio features a wide array of cardiac interventions, with a specific emphasis on engineering complex fixed-dose combinations (FDCs) that enhance patient compliance by combining multiple active ingredients into a single, highly stable tablet.²⁷ The firm also maintains a robust pipeline in Diabetes and Endocrinology, aiming for a leadership position by developing the complete spectrum of diabetic management tools, particularly focusing on sustained-release architectures that prevent dangerous glycemic spiking.²⁷ Furthermore, the company develops extensive product lines spanning Urology, Gastroenterology, Pain Management, Anti-infectives, and complex Dermatological Topicals.²⁷

3.2 Landmark Approvals: Methenamine Hippurate and Complex Generics

The commercial viability of Lyrus's R&D engine is evidenced by several highly significant regulatory triumphs in the most competitive global markets. The paramount achievement in the company's recent history occurred in December 2023, when the United States Food and Drug Administration (US FDA) formally approved Lyrus's Abbreviated New Drug Application (ANDA) for Methenamine Hippurate Tablets USP, 1 gram.¹¹

This approval represents an AB-rated, fully substitutable generic equivalent of the reference listed drug Hiprex®, originally marketed by Validus Pharmaceuticals LLC.¹¹ Methenamine Hippurate is a highly critical urological intervention indicated for the prophylactic and suppressive treatment of frequently recurring urinary tract infections (UTIs) in patients where long-term therapeutic suppression is deemed clinically necessary. The strategic value of this specific molecule cannot be overstated. From a macroeconomic healthcare perspective, the North American and European markets are experiencing a massive surge in the consumption of Methenamine Hippurate. This is driven by aging demographics, an increase in complex

urological procedures, rising rates of postmenopausal UTIs, and, most critically, a global medical mandate to combat antimicrobial resistance (AMR) by pivoting away from conventional daily antibiotics in favor of non-antibiotic suppressive agents.¹¹ By securing a first-cycle approval for this molecule, Lyrus has positioned itself to capture immense market share. To further solidify the clinical backing of this formulation, Lyrus has engaged in structured Phase IV post-marketing surveillance—a multi-center, prospective observational study designed to continually evaluate the long-term safety and efficacy of their Methenamine formulation in real-world patient cohorts.³⁵

Simultaneously, Lyrus has established a dominant presence in the United Kingdom regarding diabetic management therapies. The company successfully developed and secured Marketing Authorizations (MAs) for Metformin Hydrochloride Extended Release (XR) tablets across the complete dosing spectrum of 500mg, 750mg, and 1000mg.²⁰ Achieving uniform release kinetics for high-dose, highly soluble molecules like Metformin is notoriously difficult, validating the advanced polymer matrix engineering capabilities of the Hoskote formulation team.

In a display of aggressive inorganic expansion, Lyrus also successfully executed the acquisition of 46 existing prescription Marketing Authorizations directly from Sandoz Australia.⁹ Following the acquisition, the Lyrus regulatory and technical teams managed the incredibly complex process of site transfer for all 46 molecules, ensuring zero disruption to the supply chain. To monetize this massive portfolio, Lyrus secured an exclusive distribution partnership with a premier regional distributor to deeply penetrate the Australia and New Zealand (ANZ) markets, alongside successfully developing and commercializing 12 bespoke OTC formulations specifically tailored for the ANZ demographic.⁹

3.3 Intellectual Property Moat and Advanced Formulation Pipeline

A core pillar of the Lyrus corporate strategy involves constructing a dense intellectual property moat to protect its NDDS innovations from generic erosion. The firm actively files and prosecutes patents across multiple global jurisdictions, primarily targeting the United States, the European Patent Office (EPO), and the South African patent registry.

Table 3 systematically outlines the publicly identifiable patent architecture and advanced formulation pipeline currently held by Lyrus Life Sciences.

Patent / Application Number	Title and Technological Focus	Jurisdictional Status and Strategic Notes
US 12318425 B2	Formulation comprising water-soluble particles of a non-curcuminoid and a curcuminoid	Granted (USA). This landmark patent details the engineering of a highly water-soluble, nanotech-driven formulation combining Turmeric oil, oleoresin, and specific curcuminoids. Crucially, the technology achieves thermodynamic stability without relying on conventional pH buffering agents, minimizing gastric distress. ³²
EP 4153146 A1 / ZA 202207992 B / US 20230158160	Pharmaceutical Composition of Extended-Release Oral Suspension and Process for Preparation Thereof	Published Applications / Grants (EU, SA, US). This intellectual property protects the complex rheological matrices required to keep active ingredients uniformly suspended in a liquid state while maintaining a strictly controlled, extended-release pharmacokinetic profile upon ingestion. ²¹
EP 4185268 A1 / ZA 202207991 B	Novel Processes for the Preparation of Rapid Melt Granules	Published Applications / Grants (EU, SA). Details advanced delivery systems engineered for instantaneous oral dispersion without the need for water, providing a critical competitive edge in pediatric and geriatric therapeutic markets. ²¹
EP 4294418	Novel Arthritis Emulgel Composition and Its	Published Application (EU). Protects a highly specialized topical application that fuses the structural stability of an emulsion with the

A1 Preparation Process thermodynamic cooling and rapid transdermal penetration properties of a gel, specifically targeted for deep-tissue arthritis relief.²¹

To further accelerate the commercialization of this pipeline, Lyrus formalized a strategic innovation collaboration in early 2023 with Akums Drugs & Pharmaceuticals, one of India's largest contract manufacturers.³⁸ This joint-development synergy allows Lyrus to overlay its advanced NDDS patents onto Akums' unparalleled mass-manufacturing infrastructure, ensuring that highly complex, patient-friendly formulations can be brought to the global market at massive scale and competitive pricing.³⁸

4. Market Positioning, B2B Export Strategy, and Competitive Dynamics

4.1 The B2B Out-Licensing Strategy and Extreme Client Concentration

Unlike vertically integrated pharmaceutical corporations that manage the entire process from bench research to retail pharmacy shelves, Lyrus Life Sciences operates via a highly focused, Business-to-Business (B2B) out-licensing and supply methodology.⁵ The company rarely engages directly with patients or healthcare providers. Instead, the business model is predicated on developing highly complex regulatory dossiers and subsequently out-licensing the marketing rights, or acting as the exclusive contract manufacturer for global pharmaceutical giants, mid-tier regional distributors, and prominent retail pharmacy chains.⁵

To facilitate the friction-free transfer of intellectual property and manage multi-national taxation and regulatory requirements, Lyrus operates a network of international subsidiaries. The company directs its European operations through Lyrus Life Sciences Limited, headquartered in Uxbridge, United Kingdom, and manages its North American strategic interests via Lyrus Life Sciences Inc., incorporated in Wilmington, Delaware.²⁰

However, an analysis of the corporate revenue streams reveals a state of extreme customer concentration. According to the FY23 financial reviews, the top four corporate clients accounted for an astonishing 85% of Lyrus's total sales volume.⁵ More specifically, a single multinational entity—the Noumed Group, comprising Noumed Pharmaceuticals and Noumed Life Sciences—stood as the undisputed keystone of the Lyrus commercial enterprise, representing approximately 68% of total gross sales.⁵

An analysis of global export logs and regulatory filings illuminates the depth of this partnership. Lyrus acts as the primary manufacturing and formulation partner for vast quantities of Noumed-branded therapeutics, exporting millions of units of Paracetamol 500mg tablets, Cetirizine 10mg allergy relief tablets, Fexofenadine 120mg film-coated tablets, and the critical anti-hypertensive Lisinopril 5mg.³⁶ Beyond the Noumed relationship, Lyrus also maintains deep commercial ties with Zentiva Pharma UK, supplying the aforementioned Metformin XR formulations and various complex generics.³⁶ While relying on a single client for nearly 70% of revenue presents a profound theoretical risk to corporate continuity, this vulnerability is practically mitigated by the exceptionally strong credit profile, immense market capitalization, and entrenched supply chain stability of the Noumed Group within the UK and Australian healthcare systems.⁵

4.2 Global Export Corridors and Inorganic Market Penetration

The geographic commercial focus of Lyrus Life Sciences is overwhelmingly directed toward highly regulated and semi-regulated export markets, consciously bypassing the hyper-competitive and price-sensitive domestic Indian generic landscape.⁵ The primary export corridors currently heavily trafficked by the organization include the United Kingdom, the broader European Union, Australia, and New Zealand.⁴ Having secured foundational footholds in these tier-one regulatory environments, Lyrus has mapped an aggressive secondary expansion strategy. The executive mandate involves leveraging the clinical prestige and rigorous data packages of its US FDA and EMA approvals to monetize emerging markets without

requiring ground-up clinical trials. Specifically, Lyrus is executing strong localized distribution partnerships to penetrate the vast Sub-Saharan African, Middle Eastern, North African (MENA), and Latin American (LATAM) healthcare sectors.⁹

4.3 Competitor Landscape in the CDMO/CRDO Sector

The global and Indian Contract Development and Manufacturing Organization (CDMO) and CRDO sectors represent a highly saturated, multi-billion-dollar battleground. While macro-level corporate databases suggest Lyrus contends with over 870 active competitors, a granular analysis of its specific operational niche — focusing on high-barrier complex generics, advanced nanotech NDDS, and dossier out-licensing — significantly refines the true competitor field.⁶

Table 4 systematically analyzes the primary strategic competitors and their respective market postures relative to the operational model of Lyrus Life Sciences.

Competitor Entity	Corporate Status / Headquarters	Market Overlap and Competitive Posture
Rubicon Research	Funded Private Entity (India)	A highly capitalized, direct CDMO competitor. Rubicon focuses aggressively on complex generics, 505(b)(2) regulatory pathways, and advanced formulation science for highly regulated Western markets. It is highly notable that Lyrus's current Head of R&D, Elayaraja Natarajan, is a former executive at Rubicon, indicating a direct transfer of strategic institutional knowledge between the competing firms. ⁶
Akums Drugs & Pharmaceuticals	Public Entity (India)	Historically a dominant competitor in mass-scale Indian contract manufacturing. However, the competitive dynamic shifted radically in 2023 when Lyrus pivoted to form a strategic joint-development alliance with Akums, effectively transitioning a massive competitor into a highly synergistic manufacturing partner. ⁶
Jubilant Life Sciences	Public Multinational (India)	A massive global conglomerate that competes broadly in the overarching CRDO space. However, Jubilant focuses heavily on active pharmaceutical ingredients (APIs), radiopharmaceuticals, and high-volume generic solids, presenting a less direct threat to Lyrus's highly specialized NDDS out-licensing model. ⁶
Natco Pharma	Public Entity (India)	Renowned globally for aggressive patent litigation (Paragraph IV certifications) and the engineering of highly complex oncology and specialty generic therapeutics. Natco acts as a formidable competitor to Lyrus in the US ANDA space, particularly concerning first-to-file applications and niche formulation products. ⁶
Adcock Ingram	Public Entity (South Africa)	While tracing its roots back to 1890, Adcock dominates the local generic and specialty distribution networks in Sub-Saharan Africa. As Lyrus executes its strategy to penetrate the African continent, it must directly compete with Adcock's entrenched distribution supremacy and localized manufacturing dominance. ⁶
Supriya Lifescience	Public Entity (India)	Primarily operates as a manufacturer of APIs rather than finished dosages, meaning it competes adjacently to Lyrus within the broader supply chain ecosystem without directly encroaching on Lyrus's formulation dominance. ⁶

5. Operational Bottlenecks and Digital Transformation Opportunities

Despite the immense scientific triumphs achieved by the Hoskote R&D facility and the successful penetration of highly regulated global markets, the aggressive expansion of Lyrus Life Sciences has exposed several profound structural and operational bottlenecks. These pain points are endemic to mid-sized, rapidly scaling CRDOs that must bridge the perilous gap between capital-intensive clinical research and profitable commercialization. A rigorous analysis of these operational friction points reveals highly lucrative opportunities for sweeping digital modernization, process automation, and the integration of Artificial Intelligence (AI) architectures.

5.1 Diagnosing Crucial Operational Pain Points

1. The Milestone Billing Trap and the Bloated Gross Current Asset (GCA) Cycle: The single most severe operational constraint threatening the financial fluidity of Lyrus Life Sciences is its revenue recognition architecture. The contract R&D division invoices its massive international clientele—such as the Noumed Group and Zentiva—based on a stringent milestone billing model.⁵ Within the pharmaceutical development cycle, a single project is typically gated behind massive milestones: successful pilot batch stability, completion of pivotal in-vivo bioequivalence studies, and the ultimate filing of the regulatory dossier. Achieving any single milestone can require six to eighteen months of intensive labor.

During this protracted period, Lyrus assumes immense, continuous financial liabilities. The firm must fund the salaries of 65+ highly specialized scientists, procure exorbitantly expensive reference APIs and analytical reagents, and maintain the immense overhead of the Hoskote cGMP facility. Because Lyrus cannot legally invoice the client until the milestone is formally achieved, documented, and approved, the enterprise generates a massive backlog of unbilled receivables. This phenomenon traps working capital and entirely strangulates operating cash flow, directly resulting in an extraordinarily bloated Gross Current Asset (GCA) cycle. Historical financial data indicates this GCA cycle stretched to a staggering 424.61 days in FY23 (Provisional).⁵ This 14-month cash flow delay is the exact mechanism that forces the continual, expensive infusion of external debt and promoter equity merely to sustain baseline operations.⁵

2. Extreme Product and Customer Concentration Vulnerability: As established, approximately 68% of corporate sales stem from a single entity (Noumed), with the top four clients comprising 85% of total revenue.⁵ Compounding this customer concentration is an equally severe product concentration risk. The vast majority of the company's highly profitable traded pharmaceutical revenue is driven by a single molecule class: high-volume Paracetamol-related products exported to Australia and the UK.⁵ If the Noumed Group were to pivot its supply chain to an alternative CDMO, or if global Paracetamol API prices were to experience extreme volatility—a frequent occurrence driven by fluctuations in Chinese chemical manufacturing—Lyrus's primary revenue pillar would face acute, existential threat.⁵

3. Unhedged Foreign Exchange (FX) Exposure: Because Lyrus operates almost exclusively as an exporter to the UK, Australia, Europe, and the United States, its entire revenue model is fundamentally linked to global currency valuations.⁵ Shockingly, financial audits reveal that the company operates without a formalized, institutional currency hedging mechanism. By keeping its export leg entirely open and booking complex, multi-million dollar transactions at the unpredictable daily spot rate, Lyrus effectively gambles its profit margins on macroeconomic stability.⁵ In FY23 alone, the company carried an unhedged foreign exposure totaling roughly ₹30 Crore.⁵ While macroeconomic luck has allowed the firm to realize FX gains in recent cycles, operating without a rigorous hedging policy represents an unacceptable structural risk for an enterprise dealing with thin generic manufacturing margins.

4. The Friction of Regulatory Dossier Compilation and QA/QC Workflows: The lifeblood of any CRDO is the generation of irreproachable, mathematically flawless data. When Lyrus compiles a highly complex ANDA for the US FDA or an MA for the MHRA, the sheer volume of required documentation is staggering.¹¹ A single dossier encompasses thousands of pages detailing pre-formulation thermodynamic stability data, analytical method validations, clinical bioequivalence metrics, and highly specific Chemistry, Manufacturing, and Controls (CMC) data. In conventional, siloed IT environments that rely on manual data transcription, the

Quality Assurance and Quality Control (QA/QC) review workflows become notoriously slow and highly prone to human error. Every day spent manually reviewing chromatography data delays the regulatory submission, which subsequently delays the milestone billing trigger, directly feeding back into the catastrophic 424-day GCA cycle.

5.2 Disambiguation of Publicly Known Software Systems

Before prescribing technological solutions, it is imperative to address a significant point of potential industry confusion regarding the software nomenclature associated with the company's name. In the domains of computational biology, genomics, and bioinformatics, there exists a highly publicized, open-source machine learning model programmed in Python simply titled "LYRUS".⁴³ This academic software utilizes advanced algorithms and FoldX energetics to predict the clinical pathogenicity of single amino acid variants (SAVs) and missense mutations based on experimental protein structures (such as PTEN and TP53).⁴⁴

It must be explicitly stated that there is zero evidence connecting this advanced genomic prediction tool to the internal corporate IT infrastructure of Lyrus Life Sciences Private Limited. As a cGMP-compliant formulation manufacturer, Lyrus Life Sciences does not engage in de novo genomic sequencing. Instead, the Hoskote facility relies on highly regulated, 21 CFR Part 11 compliant enterprise software. This ecosystem undoubtedly includes conventional Enterprise Resource Planning (ERP) frameworks for inventory management, high-performance chromatography data systems (CDS) for analytical instrumentation, and robust Laboratory Information Management Systems (LIMS) for batch record compliance.

5.3 High-ROI Digital Interventions and AI Integration Architectures

Based on the specific operational architecture and diagnosed pain points of Lyrus Life Sciences, the strategic deployment of targeted digital platforms and AI-driven agents would yield immediate, transformative Returns on Investment (ROI):

1. AI-Powered Regulatory Dossier Automation (To Collapse the GCA Cycle):

The most direct and urgent mechanism to reduce the crippling 424-day GCA cycle is to exponentially accelerate milestone completion. Lyrus must implement specialized Natural Language Processing (NLP) agents and fine-tuned Large Language Models (LLMs) trained exhaustively on ICH guidelines, eCTD (Electronic Common Technical Document) formatting rules, and FDA/EMA submission templates. These AI agents can be engineered to automatically ingest raw numerical outputs from laboratory instruments—such as the complex dynamic light scattering data generated by the Brookhaven Nano analyzer—and autonomously draft the highly technical CMC sections of the ANDA/MA dossiers. By reducing the time required for manual data transcription, QA/QC peer review, and final dossier formatting by even 30 to 40%, Lyrus could achieve its contractual milestones months earlier, radically improving working capital liquidity and reducing the reliance on high-interest debt.

2. Predictive AI Hedging and Automated Financial Treasury Operations: Given the severe structural risk posed by the ₹30 Crore unhedged foreign exposure, the immediate implementation of an automated, AI-driven financial treasury platform is imperative.⁵ A predictive machine learning hedging agent could constantly analyze historical cash flow requirements, predicted milestone completion timelines, and real-time macroeconomic indicators across the INR/GBP, INR/AUD, and INR/USD corridors. The agent would dynamically recommend—and autonomously execute—optimal forward contracts or currency options. This algorithmic shield would mathematically isolate Lyrus's razor-thin export margins from the volatility of global monetary policy, completely neutralizing the current unhedged risk.

3. Next-Generation LIMS for Nanotech and Formulation Tracking: Lyrus's pioneering work in nanonization—specifically the complex engineering required to reduce curcuminoid particles to the 40-157 nm range—generates massive, highly complex datasets regarding rheology, dissolution kinetics, and real-time stability metrics under accelerated thermodynamic conditions.³⁰ Upgrading the Hoskote R&D center to a modernized, cloud-native Laboratory Information Management System (LIMS) fully integrated with deep-

learning data mining capabilities is essential. This would allow the Ideation and Formulation teams to track and correlate data across thousands of historical experimental iterations. By utilizing a "digital twin" approach to pharmaceutical formulation, the AI could predict API-excipient incompatibilities and thermodynamic instabilities entirely in silico before physical compounding begins. This would save massive amounts of R&D capital, drastically reduce the failure rate of physical pilot batches, and accelerate the speed to market for novel NDDS platforms.

4. Modernized B2B Client Portal for Unprecedented Milestone Transparency:

Currently, client reporting and milestone tracking in mid-sized CRDOs are often handled via archaic, fragmented methods involving periodic email updates and static spreadsheet exchanges. By engineering and deploying a secure, hyper-modernized B2B web portal, Lyrus could provide real-time, interactive dashboards to its most critical clients, namely the Noumed Group and Zentiva. This portal would visualize the exact, daily progress of formulation development, ongoing long-term stability studies, and regulatory milestone achievements. This level of extreme transparency achieves three massive ROI objectives: it fortifies unparalleled client trust, it heavily mitigates the Noumed concentration risk by embedding Lyrus's digital infrastructure deeply into the client's own supply chain software ecosystem, and it enables instantaneous, automated invoicing the very second a digital milestone is mutually approved on the platform, further collapsing the GCA cycle and securing the financial future of the enterprise.

Works cited

1. About Lyrus Life Sciences | CPHI Online, accessed May 21, 2026, <https://www.cphi-online.com/company/lyrus-life-sciences/> (<https://www.cphi-online.com/company/lyrus-life-sciences/>)
2. About Us - Trust Pharmacy, accessed May 21, 2026, <https://www.trustpharmacy.co.in/About.html> (<https://www.trustpharmacy.co.in/About.html>)
3. Medreich Limited - ICRA, accessed May 21, 2026, https://www.icra.in/Rating/GetRationalReportFilePdf/12838~Medreich%20Limited_r_04082015.pdf (https://www.icra.in/Rating/GetRationalReportFilePdf/12838~Medreich%20Limited_r_04082015.pdf)
4. About Us - veristonN, accessed May 21, 2026, <https://veristonn.com/about.html> (<https://veristonn.com/about.html>)
5. Lyrus Life Sciences Private Limited July 06, 2023, accessed May 21, 2026, https://infomericstorage.blob.core.windows.net/uploads/pr_Lyrus_Life_Sciences_6jul23_83543c38a/ (https://infomericstorage.blob.core.windows.net/uploads/pr_Lyrus_Life_Sciences_6jul23_83543c38a/)
6. Lyrus Life Sciences - 2026 Company Profile, Team, Competitors & Financials - Tracxn, accessed May 21, 2026, https://tracxn.com/d/companies/lyruslifesciences/_JFm9dDdXJeiTaxS-lqCKyoQPv-oGuUsnaC-ZBPBY7yc (https://tracxn.com/d/companies/lyruslifesciences/_JFm9dDdXJeiTaxS-lqCKyoQPv-oGuUsnaC-ZBPBY7yc)
7. Lyrus Life Sciences Private Limited Company Profile - India | Financials & Key Executives, accessed May 21, 2026, https://www.emis.com/php/company-profile/IN/Lyrus_Life_Sciences_Private_Limited_en_6839425.html (https://www.emis.com/php/company-profile/IN/Lyrus_Life_Sciences_Private_Limited_en_6839425.html)
8. Lyrus Life Sciences Private Limited - U24230KA2015PTC080902 - InstaFinancials, accessed May 21, 2026, <https://www.instafinancials.com/company/lyrus-life-sciences-private-limited-U24230KA2015PTC080902> (<https://www.instafinancials.com/company/lyrus-life-sciences-private-limited-U24230KA2015PTC080902>)
9. Lyrus Life Sciences - CPHI Online, accessed May 21, 2026, <https://www.cphi-online.com/brochure/lyrus-life-science-brochure/Lyrus%20Corporate%20Presentation.pdf> (<https://www.cphi-online.com/brochure/lyrus-life-science-brochure/Lyrus%20Corporate%20Presentation.pdf>)
10. Generic and Specialty Formulations - Lyrus Life Sciences, accessed May 21, 2026, <https://www.lyruslife.com/generic-and-specialty-formulations/> (<https://www.lyruslife.com/generic-and-specialty-formulations/>)

[and-specialty-formulations/](#)

11. Lyrus Receives US FDA Approval for Critical UTI Treatment in the US - ANI News, accessed May 21, 2026, <https://www.aninews.in/news/business/business/lyrus-receives-us-fda-approval-for-critical-uti-treatment-in-the-us20231215110816/>
(<https://www.aninews.in/news/business/business/lyrus-receives-us-fda-approval-for-critical-uti-treatment-in-the-us20231215110816/>)
12. Lyrus Life Sciences in Chokkahalli, Bangalore - Justdial, accessed May 21, 2026, https://www.justdial.com/Bangalore/Lyrus-Life-Sciences-Chokkahalli/080PXX80-XX80-221019230129-D7I3_BZDET (https://www.justdial.com/Bangalore/Lyrus-Life-Sciences-Chokkahalli/080PXX80-XX80-221019230129-D7I3_BZDET)
13. Mr. C. P. Bothra - Stabicon - Incisive Solution, Decisive Results, accessed May 21, 2026, <https://stabicon.com/bothra.php> (<https://stabicon.com/bothra.php>)
14. accessed May 21, 2026, <https://stabicon.com/bothra.php#:~:text=Mr.-.C.%20P.,co%20wner%20of%20Medreich%20Limited.>
(<https://stabicon.com/bothra.php#:~:text=Mr.-.C.%20P.,co%20wner%20of%20Medreich%20Limited.>)
15. Trust Pharmacy, accessed May 21, 2026, <https://www.trustpharmacy.co.in/>
(<https://www.trustpharmacy.co.in/>)
16. Leela Kanwar (DIN: 07110887) - Partner & Director at Equidor Medtech Llp & 5 More - CompanyRadar, accessed May 21, 2026, <https://www.companyradar.in/people/leela-kanwar/07110887?tab=overview> (<https://www.companyradar.in/people/leela-kanwar/07110887?tab=overview>)
17. Lyrus Life Sciences Private Limited - CARE Ratings, accessed May 21, 2026, https://www.careratings.com/upload/CompanyFiles/PR/18072022061103_Lyrus_Life_Sciences_Priv
(https://www.careratings.com/upload/CompanyFiles/PR/18072022061103_Lyrus_Life_Sciences_Priv)
18. LEELA KANWAR I DIN 07110887 Director Profile - IndiaFilings, accessed May 21, 2026, <https://www.indiafilings.com/search/leela-kanwar-din-07110887>
(<https://www.indiafilings.com/search/leela-kanwar-din-07110887>)
19. Crossings - Issue 03 - August 2022.cdr - Rotary India, accessed May 21, 2026, https://rotaryindia.org/Documents/ebulletin/Group1579/Crossings_-_Issue_03_-_August_202209082022102424AM.pdf
(https://rotaryindia.org/Documents/ebulletin/Group1579/Crossings_-_Issue_03_-_August_202209082022102424AM.pdf)
20. MRS. UMA MAHESHWARI LYRUS LIFE SCIENCES LIMITED C/O THAKUR-CHABERT, SUITE 410, 4TH FLOOR 1 HAREFIELD ROAD UXBRIDGE UB8 1EX UNITE - CPHI Online, accessed May 21, 2026, <https://www.cphi-online.com/news/mhra-approvals/metformin-xr-1000mg.pdf>
(<https://www.cphi-online.com/news/mhra-approvals/metformin-xr-1000mg.pdf>)
21. Lyrus life sciences pvt ltd Patents - PatentGuru, accessed May 21, 2026, <https://www.patentguru.com/assignee/lyrus-life-sciences-pvt-ltd>
(<https://www.patentguru.com/assignee/lyrus-life-sciences-pvt-ltd>)
22. Lyrus Life Sciences Financials I Company Details - Tofler, accessed May 21, 2026, <https://www.tofler.in/lyrus-life-sciences-private-limited/company/U24230KA2015PTC080902>
(<https://www.tofler.in/lyrus-life-sciences-private-limited/company/U24230KA2015PTC080902>)
23. LYRUS LIFE SCIENCES PRIVATE LIMITED - 2026 Company Profile & Financials - Tracxn, accessed May 21, 2026, https://tracxn.com/d/legal-entities/india/lyrus-life-sciences-private-limited/_lp7iakupLVDMCS3oCpxdBvor-G_XG3tdFkaYxq9Y3iY (https://tracxn.com/d/legal-entities/india/lyrus-life-sciences-private-limited/_lp7iakupLVDMCS3oCpxdBvor-G_XG3tdFkaYxq9Y3iY)
24. Lyrus Life Sciences Private Limited (LLSPL) July 11, 2024, accessed May 21, 2026, https://infomericstorage.blob.core.windows.net/uploads/pr_LLSPL_11july24_30b5f5ef0d.pdf
(https://infomericstorage.blob.core.windows.net/uploads/pr_LLSPL_11july24_30b5f5ef0d.pdf)
25. Lyrus Life Science Pvt. Ltd. - ISTI Portal, accessed May 21, 2026, <https://istidelhi.php->

- staging.com/organisations/st-organisations-across-country/2123-lyrus-life-science-pvt-ltd
(<https://istidelhi.php-staging.com/organisations/st-organisations-across-country/2123-lyrus-life-science-pvt-ltd>)
26. Company Profile - Lyrus Life Sciences Pvt Ltd - ChemDmart, accessed May 21, 2026, <https://chemdmart.com/company-profile/lyrus-life-sciences-pvt-ltd>
(<https://chemdmart.com/company-profile/lyrus-life-sciences-pvt-ltd>)
 27. Lyrus Life Sciences: Home, accessed May 21, 2026, <https://www.lyruslife.com/>
(<https://www.lyruslife.com/>)
 28. Manufacturing - Lyrus Life Sciences, accessed May 21, 2026, <https://www.lyruslife.com/manufacturing/> (<https://www.lyruslife.com/manufacturing/>)
 29. Pilot and Exhibit Manufacturing - Lyrus Life Sciences, accessed May 21, 2026, <https://www.lyruslife.com/pilot-and-exhibit-manufacturing/> (<https://www.lyruslife.com/pilot-and-exhibit-manufacturing/>)
 30. US20220054580A1 - Formulation comprising water soluble particles of a non-curcuminoid ... - Google Patents, accessed May 21, 2026, <https://patents.google.com/patent/US20220054580A1/en>
(<https://patents.google.com/patent/US20220054580A1/en>)
 31. WO2020084634A1 - Formulation comprising water soluble particles of a non-curcuminoid ... - Google Patents, accessed May 21, 2026, <https://patents.google.com/patent/WO2020084634A1/en>
(<https://patents.google.com/patent/WO2020084634A1/en>)
 32. Patents Assigned to LYRUS LIFE SCIENCES PVT. LTD., accessed May 21, 2026, <https://patents.justia.com/assignee/lyrus-life-sciences-pvt-ltd>
(<https://patents.justia.com/assignee/lyrus-life-sciences-pvt-ltd>)
 33. Lyrus Life Sciences Pvt. Ltd. in Bengaluru, Karnataka, India - Company Profile - Tradeindia, accessed May 21, 2026, <https://www.tradeindia.com/lyrus-life-sciences-pvt-ltd-19086159/>
(<https://www.tradeindia.com/lyrus-life-sciences-pvt-ltd-19086159/>)
 34. Human Health Care - Nutraceutical & Standardized Herbal Extracts, accessed May 21, 2026, <https://www.indiamart.com/lyrus-life-sciences-private-limited/human-health-care-nutraceutical-standardized-herbal-extracts.html> (<https://www.indiamart.com/lyrus-life-sciences-private-limited/human-health-care-nutraceutical-standardized-herbal-extracts.html>)
 35. Recommendations of the SEC (Urology) made in its 09th/25 meeting held on 13.11.2025 at - CDSCO, accessed May 21, 2026, [https://cdsco.gov.in/opencms/resources/UploadCDSCOWeb/2018/UploadCommitteeFiles/Recommendations%20of%20the%20SEC%20\(Urology\)%20made%20in%20its%2009th%20meeting%20held%20on%2013.11.2025%20at%20CDSCO.pdf](https://cdsco.gov.in/opencms/resources/UploadCDSCOWeb/2018/UploadCommitteeFiles/Recommendations%20of%20the%20SEC%20(Urology)%20made%20in%20its%2009th%20meeting%20held%20on%2013.11.2025%20at%20CDSCO.pdf)
([https://cdsco.gov.in/opencms/resources/UploadCDSCOWeb/2018/UploadCommitteeFiles/Recommendations%20of%20the%20SEC%20\(Urology\)%20made%20in%20its%2009th%20meeting%20held%20on%2013.11.2025%20at%20CDSCO.pdf](https://cdsco.gov.in/opencms/resources/UploadCDSCOWeb/2018/UploadCommitteeFiles/Recommendations%20of%20the%20SEC%20(Urology)%20made%20in%20its%2009th%20meeting%20held%20on%2013.11.2025%20at%20CDSCO.pdf))
 36. Own label suppliers list, accessed May 21, 2026, <https://mhra.disclosurelog.co.uk/documents/9a781ef5-7a28-4888-8ca9-ec9fd0d1eb7d>
(<https://mhra.disclosurelog.co.uk/documents/9a781ef5-7a28-4888-8ca9-ec9fd0d1eb7d>)
 37. Marketing authorisations granted 1 June 2024 to 14 June 2024 - GOV.UK, accessed May 21, 2026, https://assets.publishing.service.gov.uk/media/66963abcab418ab0555925e6/Marketing_authorisations_granted_1_June_2024_to_14_June_2024.pdf
(https://assets.publishing.service.gov.uk/media/66963abcab418ab0555925e6/Marketing_authorisations_granted_1_June_2024_to_14_June_2024.pdf)
 38. Akums, Lyrus Life Sciences collaborate for patented technologies and formulations, accessed May 21, 2026, <https://health.economictimes.indiatimes.com/news/pharma/akums-lyrus-life-sciences-collaborate-for-patented-technologies-and-formulations/98148284>
(<https://health.economictimes.indiatimes.com/news/pharma/akums-lyrus-life-sciences-collaborate-for-patented-technologies-and-formulations/98148284>)
 39. Contact Us - Lyrus Life Sciences, accessed May 21, 2026, <https://www.lyruslife.com/contact-us/>
(<https://www.lyruslife.com/contact-us/>)
 40. LYRUS LIFE SCIENCES LIMITED overview - Find and update company information, accessed May 21, 2026, <https://find-and-update.company-information.service.gov.uk/company/10436822>
(<https://find-and-update.company-information.service.gov.uk/company/10436822>)
 41. Lyrus Life Sciences Private Limited Import Export Data & Shipment Details - eximpedia, accessed May 21, 2026, <https://www.eximpedia.app/companies/lyrus-life-sciences-private-limited/03652557>
(<https://www.eximpedia.app/companies/lyrus-life-sciences-private-limited/03652557>)

42. Lyrus Life Sciences Private Limited — Pharmaceutical Exporter, accessed May 21, 2026, <https://transdatanexus.com/supplier-profile/lyrus-life-sciences-private-limited/>
(<https://transdatanexus.com/supplier-profile/lyrus-life-sciences-private-limited/>)
43. jiaying2508/LYRUS - GitHub, accessed May 21, 2026, <https://github.com/jiaying2508/LYRUS>
(<https://github.com/jiaying2508/LYRUS>)
44. Supplemental Information for “LYRUS: A Machine Learning Model for Predicting the Pathogenicity of Missense Variants”, accessed May 21, 2026, <https://academic.oup.com/bioinformaticsadvances/article-pdf/2/1/vbab045/47085786/vbab045.pdf>
(<https://academic.oup.com/bioinformaticsadvances/article-pdf/2/1/vbab045/47085786/vbab045.pdf>)
45. (PDF) LYRUS: A Machine Learning Model for Predicting the Pathogenicity of Missense Variants - ResearchGate, accessed May 21, 2026, https://www.researchgate.net/publication/357338882_LYRUS_A_Machine_Learning_Model_for_Pre
(https://www.researchgate.net/publication/357338882_LYRUS_A_Machine_Learning_Model_for_Pre)
46. LYRUS: a machine learning model for predicting the pathogenicity of missense variants | Bioinformatics Advances | Oxford Academic, accessed May 21, 2026, <https://academic.oup.com/bioinformaticsadvances/article/2/1/vbab045/6483096>
(<https://academic.oup.com/bioinformaticsadvances/article/2/1/vbab045/6483096>)